

Problems related to Decision Making Under Uncertainty & Reinforcement Learning

ELG 5218 – Uncertainty Evaluation in Engineering
Measurements and Machine Learning

Instructor for this lecture: Bingze Xia, University of Ottawa

April 2026

Contents

I	Chapter 34: Decision Making Under Uncertainty	2
1	PART A: Conceptual Questions	2
2	PART B: Mathematical Derivations	3
3	PART C: Computational Problems	4
4	PART D: Analysis & Comparison	5
II	Chapter 35: Reinforcement Learning	6
5	PART A: Conceptual Questions	6
6	PART B: Mathematical Derivations	7
7	PART C: Computational Problems	7
8	PART D: Analysis & Comparison	8
9	PART E: External Resource Problems	9

How to Use This Document

Problems marked with [EXT] are adapted from external resources (textbooks, online courses). The specific source is cited for each. All other problems are original, designed to match the lecture content.

Part I

Chapter 34: Decision Making Under Uncertainty

1 PART A: Conceptual Questions

Problem A1: Frequentist vs. Bayesian Risk

Explain the key difference between frequentist risk and Bayesian posterior risk. In what sense does each approach treat the unknown state h and the data x ?

Problem A2: Loss Functions and Optimal Estimators

For each of the following loss functions, state the Bayes-optimal estimator and explain intuitively why it is optimal:

- (a) Zero-one loss: $\ell_{01}(h, a) = \mathbb{I}(h \neq a)$
- (b) Quadratic loss: $\ell_2(h, a) = (h - a)^2$
- (c) Absolute loss: $\ell_1(h, a) = |h - a|$

Problem A3: Bandits vs. MDPs

Explain the key structural difference between a contextual bandit and a Markov Decision Process (MDP). Why does this difference make MDPs fundamentally harder?

Problem A4: Value of Information

In the oil wildcatter example from the lecture, explain what the Value of Perfect Information (VPI) measures and how it is computed. Under what condition should the wildcatter perform the sounding test?

Problem A5: Discount Factor Interpretation

In an MDP with discount factor γ , explain:

- (a) What happens when $\gamma = 0$?

- (b) What happens when $\gamma = 1$ (for the 1D grid world from the lecture)?
- (c) Why is $\gamma = 0.9$ often a good practical choice?
- (d) Give the probabilistic interpretation of γ .

Problem A6: From MDP to POMDP

A Partially Observable MDP (POMDP) extends the standard MDP framework.

- (a) What is the key structural difference between an MDP and a POMDP?
- (b) Define the *belief state* b_t and explain why it is necessary.
- (c) Give a concrete example of a POMDP from robotics or autonomous systems, identifying the true state, the observation, and the source of partial observability.

2 PART B: Mathematical Derivations

Problem B1: Bellman Equation Derivation

Starting from the definition of the state-value function

$$V^\pi(s) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r_t \mid s_0 = s \right],$$

derive the Bellman expectation equation:

$$V^\pi(s) = \sum_a \pi(a|s) \left[R(s, a) + \gamma \sum_{s'} p(s'|s, a) V^\pi(s') \right].$$

Problem B2: Relationship Between V , Q , and A

Given the definitions:

$$\begin{aligned} V^\pi(s) &= \mathbb{E}_\pi[G_0 \mid s_0 = s], \\ Q^\pi(s, a) &= \mathbb{E}_\pi[G_0 \mid s_0 = s, a_0 = a], \\ A^\pi(s, a) &= Q^\pi(s, a) - V^\pi(s), \end{aligned}$$

prove that:

- (a) $V^\pi(s) = \sum_a \pi(a|s) Q^\pi(s, a)$
- (b) $\mathbb{E}_{a \sim \pi(\cdot|s)}[A^\pi(s, a)] = 0$
- (c) $Q^\pi(s, a) = R(s, a) + \gamma \sum_{s'} p(s'|s, a) V^\pi(s')$

Problem B3: Value Iteration Convergence

Show that Value Iteration is a contraction mapping. Specifically, let $\mathcal{T}$ denote the Bellman optimality operator:

$$(\mathcal{T}V)(s) = \max_a \left[R(s, a) + \gamma \sum_{s'} p(s'|s, a) V(s') \right].$$

Prove that for any two value functions V_1, V_2 :

$$\|\mathcal{T}V_1 - \mathcal{T}V_2\|_\infty \leq \gamma \|V_1 - V_2\|_\infty,$$

where $\|V\|_\infty = \max_s |V(s)|$.

3 PART C: Computational Problems

Problem C1: Computing Optimal Q-values

Consider the 1D grid world from the lecture with states $\{S_{T1}, S_1, S_2, S_3, S_{T2}\}$, where S_{T1} and S_{T2} are terminal (absorbing). The agent can move Up (toward S_{T1}) or Down (toward S_{T2}). Transitions are deterministic. Only $R(S_3, \text{Down}) = 1$ (reaching S_{T2}); all other rewards are 0.

Compute $Q^*(s, a)$ for all non-terminal state-action pairs using $\gamma = 0.9$.

Problem C2: Value Iteration by Hand

Consider an MDP with two non-terminal states $\{A, B\}$ and one terminal state T . The agent has two actions: a_1 and a_2 . The transition and reward model is:

State	Action	Next State	Prob.	Reward
A	a_1	B	1.0	0
A	a_2	T	1.0	2
B	a_1	A	0.5	0
B	a_1	T	0.5	10
B	a_2	A	1.0	1

Use $\gamma = 0.9$. Initialize $V_0(A) = V_0(B) = 0$. Perform **two iterations** of Value Iteration and determine the greedy policy after each iteration.

Problem C3: UCB Action Selection

External Source

Adapted from Sutton & Barto, *Reinforcement Learning: An Introduction* (2nd ed.), Chapter 2 exercises on exploration strategies.

[EXT] A 3-armed bandit has the following estimated means and counts after 20 total pulls:

Arm	$\hat{\mu}(a)$	$N_t(a)$	$\hat{\mu} + c\sqrt{\frac{\ln t}{N_t(a)}}$
1	2.5	12	?
2	3.0	5	?
3	1.8	3	?

Using $c = 2$ and $t = 20$, compute the UCB score for each arm and determine which arm UCB selects. Discuss why UCB might not select the arm with the highest estimated mean.

4 PART D: Analysis & Comparison

Problem D1: Policy Iteration vs. Value Iteration

Compare and contrast Policy Iteration and Value Iteration. Specifically:

- Describe the two steps of Policy Iteration.
- What is the computational bottleneck of each method?
- Which method typically converges in fewer iterations? Which is faster per iteration?

Part II

Chapter 35: Reinforcement Learning

5 PART A: Conceptual Questions

Problem A1: Planning vs. Learning

What is the fundamental difference between planning (Chapter 34) and reinforcement learning (Chapter 35)? What information does the RL agent lack compared to a planning agent?

Problem A2: Monte Carlo vs. TD Learning

Compare Monte Carlo (MC) and Temporal Difference (TD) methods along the following dimensions:

- (a) When can the value estimate be updated?
- (b) Bias–variance tradeoff
- (c) Which requires episodes to terminate?

Problem A3: On-Policy vs. Off-Policy

- (a) Define on-policy and off-policy learning.
- (b) Classify each of the following methods: SARSA, Q-learning, REINFORCE, PPO, DQN, SAC.
- (c) Why is off-policy learning important in practice?

Problem A4: The Deadly Triad

What is the “deadly triad” in reinforcement learning? Why does combining all three elements lead to instability?

Problem A5: DQN Innovations

Describe the two key innovations introduced by DQN (Mnih et al., 2015) that enabled stable training of deep Q-networks. Why is each necessary?

6 PART B: Mathematical Derivations

Problem B1: SARSA vs. Q-Learning Update Derivation

Write out the full update rules for SARSA and Q-learning. Starting from the general TD update pattern, show how the two differ in their choice of “next action” and explain what each converges to.

Problem B2: Policy Gradient Theorem

State the policy gradient theorem and show intuitively how it leads to the REINFORCE update. Given a parameterized policy $\pi_\theta(a|s)$, we define the objective $J(\pi_\theta) = \mathbb{E}_\pi[G_0]$. Derive the gradient $\nabla_\theta J(\pi_\theta)$ and explain the role of the baseline $b(s)$.

Problem B3: Double Q-Learning

Explain the maximization bias in standard Q-learning and derive the Double Q-learning update rule. Why does using two Q-functions solve the problem?

7 PART C: Computational Problems

Problem C1: Q-Learning Step-by-Step

External Source

Adapted from Sutton & Barto, *Reinforcement Learning: An Introduction* (2nd ed.), Example 6.6 (Cliff Walking concept) and Ch. 6 exercises.

[EXT] Consider the 1D grid world from the lecture. Apply Q-learning with $\gamma = 0.9$, $\eta = 1.0$ (full replacement), and initialize all $Q(s, a) = 0$.

The agent follows an ε -greedy policy with $\varepsilon = 1.0$ (fully random) and experiences the following episodes:

Episode 1: Start at S_3 , action $\downarrow$, reward 1, reach S_{T2} .

Episode 2: Start at S_2 , action $\downarrow$, reward 0, reach S_3 ; then action $\downarrow$, reward 1, reach S_{T2} .

Episode 3: Start at S_1 , action $\downarrow$, reward 0, reach S_2 ; action $\uparrow$, reward 0, reach S_1 ; action $\downarrow$, reward 0, reach S_2 ; action $\downarrow$, reward 0, reach S_3 ; action $\downarrow$, reward 1, reach S_{T2} .

Trace the Q-table after each transition.

Problem C2: SARSA vs. Q-Learning Comparison

Consider a simple 3-state MDP with states $\{A, B, \text{Cliff}\}$ where B is the goal (reward +10) and Cliff gives reward -100 . From state A , action “safe” goes to B with probability 1, and action “risky” goes to B with probability 0.8 and to Cliff with probability 0.2.

Using $\gamma = 1$ (episodic), compute:

- The Q-values that Q-learning converges to.
- The Q-values that SARSA converges to under a fixed ε -greedy policy with $\varepsilon = 0.1$.

(c) Which algorithm recommends the “risky” action? Is this appropriate?

Problem C3: TD Error Computation

External Source

Inspired by Sutton & Barto, *Reinforcement Learning: An Introduction* (2nd ed.), Exercise 6.1.

[EXT] An agent using TD(0) with $\eta = 0.1$ has the following value estimates:

$$V(A) = 5.0, \quad V(B) = 8.0, \quad V(C) = 3.0.$$

The agent is in state A , takes an action, receives reward $r = 2$, and transitions to state B . Using $\gamma = 0.9$:

- (a) Compute the TD error δ .
- (b) Compute the updated value $V(A)$.
- (c) If the true value $V^*(A) = 10.0$, is the update moving in the right direction?

8 PART D: Analysis & Comparison

Problem D1: Actor-Critic Architecture

Explain the Actor-Critic architecture. What does each component learn, and how do they interact? Why is this considered “the best of both worlds”?

Problem D2: PPO Clipping Mechanism

Explain how PPO’s clipped objective prevents destructive large policy updates. Given the PPO objective:

$$J^{\text{PPO}}(\pi') = \mathbb{E} \left[\min \left(\frac{\pi'(a|s)}{\pi(a|s)} A^\pi, \text{clip} \left(\frac{\pi'(a|s)}{\pi(a|s)}, 1 - \varepsilon, 1 + \varepsilon \right) A^\pi \right) \right],$$

analyze what happens for (a) a good action ($A^\pi > 0$) and (b) a bad action ($A^\pi < 0$) when the ratio $r(\theta) = \pi'(a|s)/\pi(a|s)$ grows large.

Problem D3: SAC Entropy Bonus

Soft Actor-Critic (SAC) augments the standard RL objective with an entropy term:

$$J^{\text{SAC}}(\theta) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t (R(s_t, a_t) + \lambda \mathcal{H}(\pi_\theta(\cdot|s_t))) \right].$$

- (a) What does the entropy term $\mathcal{H}(\pi_\theta(\cdot|s))$ encourage?

- (b) What happens as $\lambda \rightarrow 0$?
- (c) Why does SAC use twin critics?

Problem D4: Case Study — Algorithm Selection for UAV Pursuit-Evasion

Consider a UAV pursuit-evasion scenario: an autonomous pursuer drone must intercept an evading intruder in 3D airspace. The pursuer outputs continuous thrust and heading commands, has access only to noisy radar and LiDAR observations, and each real-world flight is expensive.

Argue why SAC is a more suitable choice than PPO or DQN for this problem. Structure your answer around three axes:

- (a) Action space
- (b) Sample efficiency
- (c) Partial observability and exploration

9 PART E: External Resource Problems

Problem E1: Bellman Equation for q_* [EXT]

External Source

Adapted from Sutton & Barto, *Reinforcement Learning: An Introduction* (2nd ed.), Exercise 3.20.

Give an equation for q_* in terms of v_* and the four-argument dynamics function $p(s', r|s, a)$.

Problem E2: Why Does TD(0) Have Lower Variance? [EXT]

External Source

Adapted from Sutton & Barto, *Reinforcement Learning: An Introduction* (2nd ed.), Chapter 6 discussion, and Murphy, *Probabilistic Machine Learning: Advanced Topics*, Section 35.2.

The MC target is $G_t = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots$. The TD(0) target is $r_t + \gamma V(s_{t+1})$. Explain mathematically why TD(0) has lower variance.

Problem E3: Exploration in Continuous Spaces [EXT]

External Source

Adapted from Lillicrap et al., *Continuous Control with Deep Reinforcement Learning* (DDPG, 2016) and Haarnoja et al., *Soft Actor-Critic* (2018).

DDPG uses a deterministic policy $a = \mu_\theta(s)$. How does it explore? Compare this to SAC's approach to exploration in continuous action spaces.

Problem E4: When to Use What? [EXT]

External Source

Based on the method selection table from Murphy, *Probabilistic Machine Learning: Advanced Topics*, Table 35.1, and the lecture summary slide.

For each of the following scenarios, recommend an RL algorithm and justify your choice:

- (a) Atari games (discrete actions, pixel observations)
- (b) Robotic arm control (continuous actions, limited real-world data)
- (c) Fine-tuning a large language model from human feedback
- (d) A board game with a known simulator (e.g., Go)

Problem E5: ε -Greedy vs. Boltzmann Exploration [EXT]

External Source

Based on Murphy, *Probabilistic Machine Learning: Advanced Topics*, Table 35.2.

Given two actions with estimated rewards $\hat{R}(s, a_1) = 4.9$ and $\hat{R}(s, a_2) = 5.1$:

- (a) Compute the selection probabilities under ε -greedy with $\varepsilon = 0.1$.
- (b) Compute the selection probabilities under Boltzmann exploration with $\tau = 1$.
- (c) Which method is more robust to estimation noise and why?

Summary of External Sources

- **Sutton & Barto**, *Reinforcement Learning: An Introduction*, 2nd edition (2018): Problems C1, C3 (Part II), E1, E2
- **Murphy**, *Probabilistic Machine Learning: Advanced Topics* (2023): Problems E4, E5
- **Lillicrap et al.** (DDPG, 2016) and **Haarnoja et al.** (SAC, 2018): Problem E3

Acknowledgements / Document Preparation

The problems and solutions in this document were generated with assistance from large language models (ChatGPT by OpenAI, CoPilot by Microsoft, and Claude by Anthropic), based on the course slide and notebook material. The instructor reviewed, modified, and verified all questions, solutions, and final formatting.